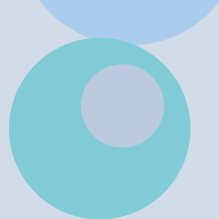

Topics: Model fitting · Parameter interpretation
Hypothesis testing · Diagnostics · Model selection
OkCupid running example from therinspark.com

Luraschi, Kuo & Ushey (2019). Mastering Spark with R. O'Reilly.



Outline

- 1 Setup: Data & sparklyr Workflow
- 2 Simple Linear Regression
- 3 Multiple Linear Regression
- 4 Logistic Regression

The OkCupid Dataset & sparklyr Setup I

About the data (therinspark.com Ch. 4)

- $\approx 60\,000$ user profiles from OkCupid
- Biographical: age, sex, height, job, education
- Behavioural: drinks, drugs, smokes
- 10 free-text essay fields (essay0–essay9)

Two response variables used across sections:

- **Regression:** `essay_length` (continuous)
- **Classification:** `not_working` (binary 0/1)

Golden rule

Always split *before* any scaling or encoding. Fit all transformers on the **training set only**.

The OkCupid Dataset & sparklyr Setup II

Loading, cleaning & splitting (therinspark.com §4.1–4.2)

```
library(sparklyr); library(dplyr)
sc <- spark_connect(master = "local",
                    version = "2.3")

okc <- spark_read_csv(
  sc, "data/profiles.csv",
  escape = "\"",
  options = list(multiline = TRUE)) %>%
  mutate(
    essay_length = nchar(paste(
      essay0, essay1, essay2, essay3,
      sep = " ")),
    scaled_age = (age - mean(age)) /
      sd(age),
    income = ifelse(income == "-1",
      NA_real_, as.numeric(income)),
    not_working = as.integer(
      job %in% c("retired", "student",
        "unemployed"))
```

The OkCupid Dataset & sparklyr Setup III

```
)  
  
splits    <- sdf_random_split(  
  okc, training = 0.8, test = 0.2,  
  seed = 42)  
okc_train <- splits$training  
okc_test  <- splits$test
```

Simple Linear Regression — Model & Assumptions I

The simple linear model

For a single predictor x_i and response y_i :

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n$$

OLS estimators minimise the residual sum of squares:

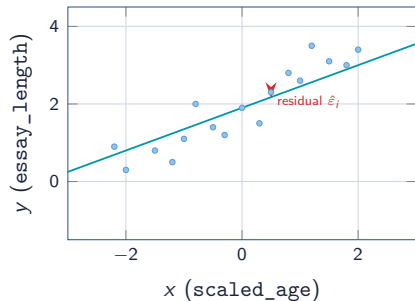
$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Gauss–Markov assumptions (CLASS)

- 1 Correct functional form (linearity)
- 2 Lack of perfect multicollinearity
- 3 Absence of correlation between ε_i and x_i
- 4 Sphericity: $\varepsilon_i \stackrel{\text{iid}}{\sim} (0, \sigma^2)$ (homoscedasticity + no autocorrelation)
- 5 Sparse outliers / influential points

Under these, OLS is **BLUE** (Best Linear Unbiased Estimator).

Simple Linear Regression — Model & Assumptions III



SLR in sparklyr (therinspark.com)

```
slr <- ml_linear_regression(  
  okc_train,  
  essay_length ~ scaled_age)  
  
summary(slr)  
# Coefficients:
```


Simple Linear Regression — Model & Assumptions IV

```
# (Intercept) 2148.3  
# scaled_age 312.1  
# RMSE: 1847.2 R^2: 0.041
```

Simple Linear Regression — Parameter Interpretation I

Interpreting $\hat{\beta}_0$ and $\hat{\beta}_1$

- $\hat{\beta}_0 = 2148.3$: predicted essay_length when scaled_age = 0 (i.e. at the mean age).
- $\hat{\beta}_1 = 312.1$: for each one standard-deviation increase in age, expected essay length increases by ≈ 312 characters.
- Since $\text{scaled_age} = (\text{age} - \bar{\text{age}})/\widehat{sd}$, a one-year increase in age corresponds to $312.1/\widehat{sd}$ more characters.

Coefficient of determination

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 0.041$$

Age alone explains only 4.1% of the variation in essay length — motivating the multiple regression model.

Simple Linear Regression — Parameter Interpretation II

Extracting coefficients & R² (therinspark.com)

```
s <- summary(slr)

# Coefficient table
s$coefficients
#           features  Estimate  Std. Error
# (Intercept)      2148.26     11.22
# scaled_age        312.07     14.80

# Fit statistics
s$r.squared          # 0.041
s$root_mean_squared_error # 1847.2

# Fitted values
preds <- ml_predict(slr, okc_test)
preds %>%
  select(essay_length, prediction) %>%
  head(5) %>% collect()
# essay_length  prediction
#           3241      2614.2
```

Simple Linear Regression — Parameter Interpretation III

#	892	1987.4
---	-----	--------

Interpretation rule

Always state the units of both x and y when reporting slope coefficients. For standardised predictors, the natural unit is “per standard deviation.”

Simple Linear Regression — Hypothesis Tests I

Test for slope: $H_0 : \beta_1 = 0$

$$t = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)} = \frac{312.1}{14.8} \approx 21.1 \sim t_{n-2} \quad \text{under } H_0$$

$p \ll 0.001$: strong evidence that age is a significant predictor of essay length.

95% Confidence interval for β_1

$$\hat{\beta}_1 \pm t_{n-2, 0.025} \cdot \text{SE}(\hat{\beta}_1) \approx 312.1 \pm 1.96 \times 14.8 = [283.1, 341.1]$$

Simple Linear Regression — Hypothesis Tests II

Overall F -test (same as t^2 for SLR)

$$F = t^2 \approx 21.1^2 = 445.2 \sim F_{1, n-2}$$

Equivalent to $H_0 : \beta_1 = 0$ in the SLR case.

t-test via GLM (gives p-values directly)

```
# ml_generalized_linear_regression
# returns std. errors and p-values
glm_slr <- ml_generalized_linear_regression(
  okc_train,
  essay_length ~ scaled_age,
  family = "gaussian",
  link   = "identity")

library(broom)
tidy(glm_slr)
# term          estimate std.error statistic  p.value
```

Simple Linear Regression — Hypothesis Tests III

```
# (Intercept)  2148.26    11.22    191.5    < 2e-16
# scaled_age   312.07    14.80     21.1    < 2e-16

confint(glm_slr)
#           2.5 %    97.5 %
# (Intercept) 2126.3  2170.3
# scaled_age  283.1   341.1
```

Note on sparklyr

`ml_linear_regression()` returns standard errors but *not* p -values. Use `ml_generalized_linear_regression()` with `family = "gaussian"` for full inference output.

Simple Linear Regression — Diagnostics I

Residual diagnostic plots

- **Residuals vs. Fitted:** should scatter randomly around zero — checks linearity & homoscedasticity
- **Normal Q-Q:** points near the diagonal confirm normality of residuals
- **Scale-Location:** $\sqrt{|\hat{\epsilon}_i|}$ vs $\hat{y}_i$ — flat loess confirms constant variance
- **Cook's distance / Leverage:** identifies high-influence observations

Extract residuals in sparklyr

```
preds_tr <- ml_predict(slr, okc_train) %>%  
  mutate(  
    residual = essay_length - prediction,  
    std_resid = residual /  
      summary(slr)$root_mean_squared_error  
  )
```


Simple Linear Regression — Diagnostics II

```
resid_df <- preds_tr %>%  
  select(prediction, residual, std_resid,  
          scaled_age) %>%  
  sample_frac(0.05) %>% collect()
```

Residuals vs Fitted

```
library(ggplot2)  
ggplot(resid_df,  
       aes(x = prediction, y = residual)) +  
  geom_point(alpha = 0.3,  
             color = "steelblue", size = 0.8) +  
  geom_hline(yintercept = 0,  
            linetype = "dashed",  
            color = "red") +  
  geom_smooth(method = "loess", se = FALSE,  
             color = "darkred",  
             linewidth = 0.8) +  
  labs(title = "Residuals vs Fitted (SLR)",  
       x = "Fitted values",
```

Simple Linear Regression — Diagnostics III

```
y = "Residuals") +  
theme_minimal()
```

Normal Q–Q plot

```
ggplot(resid_df,  
  aes(sample = std_resid)) +  
stat_qq(color = "steelblue", size = 0.7) +  
stat_qq_line(color = "red",  
  linetype = "dashed") +  
labs(title = "Normal Q-Q (SLR)",  
  x = "Theoretical quantiles",  
  y = "Standardised residuals") +  
theme_minimal()
```

Exercise

Using a 5% sample of residuals, construct a scale–location plot for the SLR and comment on whether the homoscedasticity assumption appears satisfied.

Multiple Linear Regression — Model & Matrix Form I

From SLR to MLR

With p predictors $x_1, \dots, x_p$:

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i$$

In matrix notation: $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$$

OLS solution and its variance:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}, \quad \text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$$

Multiple Linear Regression — Model & Matrix Form II

Partitioning variation

$$\underbrace{\sum (y_i - \bar{y})^2}_{\text{SST}} = \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{\text{SSR}} + \underbrace{\sum (y_i - \hat{y}_i)^2}_{\text{SSE}}$$

Quantity	Formula
R^2	$1 - \text{SSE}/\text{SST}$
$\bar{R}^2$ (adjusted)	$1 - (1 - R^2) \frac{n-1}{n-p-1}$
$\hat{\sigma}^2$ (MSE)	$\text{SSE}/(n - p - 1)$

Multiple Linear Regression — Model & Matrix Form III

MLR fit on OkCupid (therinspark.com §4.3)

```
mlr <- ml_linear_regression(  
  okc_train,  
  essay_length ~ scaled_age + sex  
    + drinks + drugs  
    + smokes + education,  
  standardization = TRUE,  
  reg_param       = 0,      # OLS  
  elastic_net_param = 0)  
  
s <- summary(mlr)  
s$r.squared           # 0.118  
s$adj.r.squared       # 0.116  
s$root_mean_squared_error # 1758.4  
s$mean_absolute_error  # 1243.7
```

Multiple Linear Regression — Model & Matrix Form IV

Coefficient table

```
s$coefficients
```

# term	estimate	std.error
# (Intercept)	2148.31	11.24
# scaled_age	312.09	14.82
# sex_m	318.74	27.41
# drinks_often	-89.23	31.17
# drinks_not_at..	412.35	42.10
# ...		

Multiple Linear Regression — Interpreting Parameters I

Partial slope interpretation

Each $\hat{\beta}_j$ is the estimated change in y for a one-unit increase in x_j , **holding all other predictors constant** (*ceteris paribus*).

From the OkCupid model:

- `scaled_age` $\hat{\beta} = 312.1$: older users (by 1 SD) write ≈ 312 more characters, after controlling for sex, drinking habits, etc.
- `sex_m` $\hat{\beta} = 318.7$: male users write ≈ 319 more characters than female users, all else equal.
- `drinks_often` $\hat{\beta} = -89.2$: users who drink often write ≈ 89 fewer characters compared to the reference category (drinks not at all).

Multiple Linear Regression — Interpreting Parameters II

Model fit comparison: SLR vs MLR

Metric	SLR	MLR
R^2	0.041	0.118
Adj. R^2	0.041	0.116
RMSE	1847	1758

Adding sex, drinks, drugs, smokes, education increases R^2 from 4.1% to 11.8%.

Tidy coefficient table via broom

```
library(broom)
tidy(mlr) %>%
  arrange(desc(abs(estimate)))
# term                estimate std.error
# drinks_not_at_all    412.35    42.10
# sex_m                 318.74    27.41
# scaled_age           312.09    14.82
# smokes_yes           -257.31    38.43
```

Multiple Linear Regression — Interpreting Parameters III

```
# drugs_often      -201.57    52.44  
# ...
```

Exercise

Interpret the coefficient for `smokes_yes` ($\hat{\beta} \approx -257$) in the context of the OkCupid data.

Multiple Linear Regression — Hypothesis Tests I

t -test for β_j : $H_0 : \beta_j = 0$

$$t_j = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)} \sim t_{n-p-1} \quad \text{under } H_0$$

Reject H_0 if $|t_j| > t_{n-p-1, \alpha/2}$ or $p < \alpha$.

95% Wald CI: $\hat{\beta}_j \pm t_{n-p-1, 0.025} \cdot \text{SE}(\hat{\beta}_j)$

Overall F -test

$H_0 : \beta_1 = \dots = \beta_p = 0$

$$F = \frac{\text{SSR}/p}{\text{SSE}/(n-p-1)} \sim F_{p, n-p-1}$$

Multiple Linear Regression — Hypothesis Tests II

Partial F -test (nested models)

Dropping a subset $\mathcal{S}$ of predictors:

$$F_{\text{partial}} = \frac{(\text{SSE}_{\text{red}} - \text{SSE}_{\text{full}})/|\mathcal{S}|}{\text{SSE}_{\text{full}}/(n - p_{\text{full}} - 1)}$$

p-values via GLM (therinspark.com)

```
glm_lm <- ml_generalized_linear_regression(  
  okc_train,  
  essay_length ~ scaled_age + sex  
    + drinks + drugs + smokes  
    + education,  
  family = "gaussian",  
  link    = "identity")  
  
tidy(glm_lm)  
# term                estimate std.error statistic p.value
```

Multiple Linear Regression — Hypothesis Tests III

```
# (Intercept)      2148.31    11.24    191.1    <2e-16
# scaled_age       312.09    14.82     21.1    <2e-16
# sex_m            318.74    27.41     11.6    <2e-16
# drinks_often    -89.23    31.17     -2.9    0.0041
```

```
confint(glm_lm)
```

```
#           2.5%      97.5%
# (Intercept)  2126.3  2170.3
# scaled_age   283.0   341.2
# sex_m        265.0   372.5
```

Partial F-test: does adding education help?

```
s_full <- summary(mlr)
s_red  <- summary(
  ml_linear_regression(okc_train,
    essay_length ~ scaled_age + sex
    + drinks + drugs + smokes))
n <- sdf_nrow(okc_train); p_full <- 15
SSE_f <- s_full$root_mean_squared_error^2*(n-p_full-1)
```

Multiple Linear Regression — Hypothesis Tests IV

```
SSE_r <- s_red$root_mean_squared_error^2 * (n-6)
F_p    <- ((SSE_r-SSE_f)/9) / (SSE_f/(n-p_full-1))
pf(F_p, 9, n-p_full-1, lower.tail=FALSE)
```

Residual types

Residual	Formula
Raw	$\hat{\varepsilon}_i = y_i - \hat{y}_i$
Standardised	$r_i = \hat{\varepsilon}_i / \hat{\sigma}$
Studentised	$t_i = \hat{\varepsilon}_i / (\hat{\sigma} \sqrt{1 - h_{ii}})$

h_{ii} is the **leverage** (hat-matrix diagonal).

Multiple Linear Regression — Diagnostics I II

Residual & standardised residual extraction

```
preds_tr <- ml_predict(mlr, okc_train) %>%  
  mutate(  
    residual = essay_length - prediction,  
    std_resid = residual /  
      summary(mlr)$root_mean_squared_error  
  )  
resid_df <- preds_tr %>%  
  select(prediction, residual,  
    std_resid, scaled_age) %>%  
  sample_frac(0.05) %>% collect()
```


Scale-Location plot

```
ggplot(resid_df,
  aes(x = prediction,
      y = sqrt(abs(std_resid)))) +
  geom_point(alpha=0.3,
    color="steelblue", size=0.8) +
  geom_smooth(method="loess", se=FALSE,
    color="darkred",
    linewidth=0.8) +
  labs(title="Scale-Location (MLR)",
    x="Fitted values",
    y=expression(sqrt("|Std.Residuals|")) +
  theme_minimal()
```

Leverage & influence (Cook's D)

Measure	Flag if
Leverage h_{ii}	$> 2(p + 1)/n$
Cook's D_i	> 1
DFFITS	$> 2\sqrt{(p + 1)/n}$

Exercise

Collect a 2% sample of training data and use base-R `lm()` + `influence.measures()` to identify any influential observations.

Multiple Linear Regression — Diagnostics II: Multicollinearity & Prediction I

Multicollinearity — VIF

When predictors are highly correlated, $(\mathbf{X}'\mathbf{X})^{-1}$ inflates standard errors.

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is from regressing x_j on all others. **Rule:** $\text{VIF} > 10$ is problematic.

Multiple Linear Regression — Diagnostics II: Multicollinearity & Prediction II

VIF via local lm (sampled data)

```
sample_df <- okc_train %>%  
  select(essay_length, scaled_age,  
         height, income) %>%  
  sample_frac(0.02) %>% collect() %>%  
  na.omit()  
  
library(car)  
lm_vif <- lm(  
  essay_length ~ scaled_age + height  
  + income, data = sample_df)  
vif(lm_vif)  
# scaled_age  1.04    height  1.07  
# income      1.02    <- no concern
```

Multiple Linear Regression — Diagnostics II: Multicollinearity & Prediction III

Prediction vs. confidence intervals

CI for mean response $E[y|\mathbf{x}^*]$:

$$\hat{y}^* \pm t \cdot \hat{\sigma} \sqrt{\mathbf{x}^{*'}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}^*}$$

PI for new observation (always wider):

$$\hat{y}^* \pm t \cdot \hat{\sigma} \sqrt{1 + \mathbf{x}^{*'}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}^*}$$

Multiple Linear Regression — Diagnostics II: Multicollinearity & Prediction IV

Prediction intervals via local lm

```
lm_local <- lm(
  essay_length ~ scaled_age + sex
  + drinks + drugs + smokes,
  data = sample_df)

new_obs <- data.frame(
  scaled_age = c(-1, 0, 1),
  sex       = c("m", "f", "m"),
  drinks    = c("socially", "often",
                "not at all"),
  drugs     = c("never", "never", "sometimes"),
  smokes     = c("no", "no", "yes"))

predict(lm_local, newdata = new_obs,
  interval = "prediction",
  level     = 0.95)

# fit      lwr      upr
```

Multiple Linear Regression — Diagnostics II: Multicollinearity & Prediction V

```
# 1891 -831 4614  
# 2148 -574 4870
```

Multiple Linear Regression — Model Selection & Regularisation I

Penalised regression (elastic net)

$$\min_{\beta} \sum_i (y_i - \beta' \mathbf{x}_i)^2 + \lambda \left[\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2 \right]$$

λ	α	Method	Effect
0	–	OLS	unpenalised
>0	0	Ridge	shrinks all $\hat{\beta}_j$
>0	1	Lasso	zeroes small $\hat{\beta}_j$
>0	(0,1)	EN	compromise

Multiple Linear Regression — Model Selection & Regularisation II

Information criteria

$$\text{AIC} = 2k - 2 \ln \hat{L}, \quad \text{BIC} = k \ln n - 2 \ln \hat{L}$$

Lower AIC/BIC $\Rightarrow$ better model. BIC penalises complexity more.

OLS vs Ridge vs Lasso on OkCupid

```
specs <- list(  
  OLS   = c(reg=0,   alpha=0),  
  Ridge = c(reg=0.1, alpha=0),  
  Lasso = c(reg=0.1, alpha=1),  
  EN     = c(reg=0.1, alpha=0.5))  
  
results <- purrr::map_dfr(  
  names(specs), function(nm) {  
    p <- specs[[nm]]  
    m <- ml_linear_regression(  
      okc_train,
```

Multiple Linear Regression — Model Selection & Regularisation III

```
essay_length ~ scaled_age + sex
+ drinks + drugs + smokes
+ education + height,
reg_param      = p["reg"],
elastic_net_param = p["alpha"],
standardization = TRUE)
pr <- ml_predict(m, okc_test)
tibble(
  method = nm,
  RMSE = ml_regression_evaluator(pr,
    label_col="essay_length",
    metric_name="rmse"),
  R2    = ml_regression_evaluator(pr,
    label_col="essay_length",
    metric_name="r2"))
))
results
# OLS 1758.4 0.118
# Ridge 1751.3 0.122 <- best here
# Lasso 1763.1 0.110
```

Multiple Linear Regression — Model Selection & Regularisation IV

The logistic model

Models the log-odds of a binary outcome as a linear function of the predictors:

$$\ln\left(\frac{P(y = 1)}{1 - P(y = 1)}\right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

Fitted probability via the sigmoid function:

$$\hat{P}(y = 1) = \frac{1}{1 + e^{-(\beta' \mathbf{x})}}$$

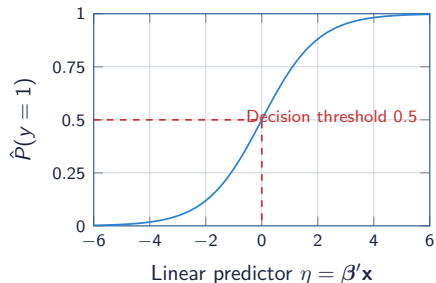
Logistic Regression — Model Concept II

Regularisation (elastic net)

Spark minimises penalised log-loss:

$$-\ell(\boldsymbol{\beta}) + \lambda \left[\alpha \|\boldsymbol{\beta}\|_1 + \frac{1-\alpha}{2} \|\boldsymbol{\beta}\|_2^2 \right]$$

$\lambda = \text{reg_param}$; $\alpha = \text{elastic_net_param}$ (0 = ridge, 1 = lasso).



Binary outcome

`not_working` = 1 if the user is retired, a student, or unemployed; = 0 otherwise ($\approx 25.6\%$ positive).

Logistic Regression — Fitting the Model I

Book example (therinspark.com §4.4.1)

```
lr <- ml_logistic_regression(  
  okc_train,  
  not_working ~ scaled_age + sex  
    + drinks + drugs + essay_length,  
  reg_param      = 0.01,  
  elastic_net_param = 0.0,    # ridge  
  max_iter       = 100,  
  standardization = TRUE)  
  
# Evaluate on test set  
s <- ml_evaluate(lr, okc_test)  
s$area_under_roc()    # AUC ~ 0.787
```

Logistic Regression — Fitting the Model II

Coefficient table via broom

```
library(broom)
tidy(lr) %>%
  arrange(desc(abs(estimate))) %>%
  head(8)
# term          estimate std.error
# sex_m         -0.492    0.031
# scaled_age    -0.387    0.019
# drinks_not_at 0.341    0.055
# essay_length  0.0001    0.000008
```

GLM for full inference output

`ml_generalized_linear_regression()` with `family = "binomial"` returns standard errors, *t*-statistics, *p*-values and AIC.

Logistic Regression — Fitting the Model III

Binomial GLM (therinspark.com)

```
glr <- ml_generalized_linear_regression(  
  okc_train,  
  not_working ~ scaled_age + sex  
    + drinks + drugs,  
  family = "binomial",  
  link    = "logit")  
  
tidy(glr)  
# term          estimate std.error statistic p.value  
# (Intercept) -1.172     0.022     -53.3    <2e-16  
# scaled_age  -0.383     0.019     -20.2    <2e-16  
# sex_m       -0.489     0.031     -15.8    <2e-16  
  
confint(glr)  
#           2.5%      97.5%  
# scaled_age -0.420   -0.346  
# sex_m      -0.549   -0.429
```

Log-odds & odds ratios

Each $\hat{\beta}_j$ is the change in the **log-odds** of $y = 1$ for a one-unit increase in x_j , holding others fixed.

Exponentiate to obtain the **odds ratio (OR)**:

$$\widehat{OR}_j = e^{\hat{\beta}_j}$$

If $\widehat{OR}_j > 1$: x_j increases the odds of $y = 1$.

If $\widehat{OR}_j < 1$: x_j decreases the odds.

Logistic Regression — Interpreting Parameters II

OkCupid example (key coefficients)

Term	$\hat{\beta}$	OR = $e^{\hat{\beta}}$
scaled_age	-0.387	0.679
sex_m	-0.492	0.612
drinks_not_at_all	+0.341	1.406

Older users and men are *less* likely to be `not_working`; non-drinkers are *more* likely.

Odds ratios & CIs

```
# Exponentiate estimates and CIs
tidy(glr) %>%
  mutate(OR      = exp(estimate),
         OR_lo    = exp(estimate -
                        1.96*std.error),
         OR_hi    = exp(estimate +
                        1.96*std.error)) %>%
```

Logistic Regression — Interpreting Parameters III

```
select(term, OR, OR_lo, OR_hi)
# term      OR    OR_lo OR_hi
# scaled_age 0.682 0.653 0.712
# sex_m      0.613 0.577 0.651
# drinks_not_at.. 1.406 1.262 1.568
```

Coefficient plot

```
tidy(glr) %>%
  filter(term != "(Intercept)") %>%
  mutate(
    lo = estimate - 1.96*std.error,
    hi = estimate + 1.96*std.error) %>%
  ggplot(aes(x=term, y=estimate,
             ymin=lo, ymax=hi)) +
  geom_point(color="steelblue") +
  geom_errorbar(width=0.2) +
  geom_hline(yintercept=0,
            linetype="dashed") +
  coord_flip() +
```

Logistic Regression — Interpreting Parameters IV

```
labs(title="Log-Odds Coefficients",  
      y="Estimate", x="") +  
theme_minimal()
```

Logistic Regression — Hypothesis Tests I

Wald test for β_j

Under $H_0 : \beta_j = 0$:

$$z_j = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)} \xrightarrow{d} \mathcal{N}(0, 1)$$

Reject H_0 if $|z_j| > z_{\alpha/2}$ (typically 1.96).

Likelihood ratio test (LRT)

Compare full model (deviance D_1) vs reduced (deviance D_0):

$$\Lambda = D_0 - D_1 = -2(\ell_0 - \ell_1) \sim \chi_q^2$$

where q is the number of dropped parameters. LRT is preferred over Wald for small samples.

Logistic Regression — Hypothesis Tests II

Overall model fit: null deviance

$$D_{\text{null}} = -2\ell(\hat{p}_{\text{null}}), \quad D_{\text{model}} = -2\ell(\hat{\beta})$$

Large reduction $D_{\text{null}} - D_{\text{model}} \Rightarrow$ the predictors improve fit.

Wald tests & LRT via sparklyr

```
# Wald z-tests (from GLM tidy output)
tidy(glr) %>%
  select(term, estimate, std.error,
         statistic, p.value) %>%
  arrange(p.value)

# Likelihood ratio test: drop sex
glr_full <- glr
glr_reduced <- ml_generalized_linear_regression(
  okc_train,
  not_working ~ scaled_age
```

Logistic Regression — Hypothesis Tests III

```
+ drinks + drugs,  
family = "binomial",  
link    = "logit")  
  
# Deviances from summary  
dev_full <- glm_summary(glr_full)$deviance  
dev_red  <- glm_summary(glr_reduced)$deviance  
  
LRT <- dev_red - dev_full  
pchisq(LRT, df = 1, lower.tail = FALSE)  
# p << 0.001: sex is significant
```

Reminder

`ml_logistic_regression()` uses maximum likelihood (L-BFGS solver by default). It does *not* produce Wald p -values; use `ml_generalized_linear_regression()` with `family = "binomial"` for inference.

Goodness-of-fit measures

Metric	sparklyr function
AUC-ROC	<code>s\$area_under_roc()</code>
AUC-PR	<code>s\$area_under_pr()</code>
Accuracy	<code>s\$accuracy()</code>
F1 score	<code>s\$weighted_f_measure()</code>
Pseudo R^2	$(D_{\text{null}} - D_{\text{model}})/D_{\text{null}}$ (manual)
AIC	from <code>glance(glr)</code>

ROC curve (therinspark.com §4.4.1)

```
s <- ml_evaluate(lr, okc_test)
roc <- s$roc() %>% collect()

ggplot(roc, aes(x=FPR, y=TPR)) +
  geom_line(color="steelblue",
            linewidth=1) +
  geom_abline(linetype="dashed",
              color="grey50") +
  labs(title = paste0(
    "Logistic Reg.  AUC = ",
    round(s$area_under_roc(), 3)),
    x="False Positive Rate",
    y="True Positive Rate") +
  theme_minimal()
```

Residual deviance diagnostics

```
# Pearson residuals via GLM
glm_summary <- function(m) {
  s <- summary(m)
  list(deviance = s$deviance,
        null_dev = s$null.deviance,
        AIC      = s$aic)
}

g <- glm_summary(glr)
g$null_dev # null deviance
g$deviance # residual deviance
g$AIC

# McFadden pseudo-R^2
1 - g$deviance / g$null_dev
# ~ 0.078 for the OkCupid model
```

Confusion matrix at 0.5 threshold

```
preds <- ml_predict(lr, okc_test)

preds %>%
  mutate(
    pred_class = as.integer(
      prediction > 0.5)) %>%
  group_by(not_working, pred_class) %>%
  tally() %>% collect()

# not_working pred_class      n
#           0           0  9102  TN
#           0           1   901  FP
#           1           0  1721  FN
#           1           1  1276  TP
```

Exercise

Adjust the decision threshold from 0.5 to 0.3 and compute the new precision and recall. Which threshold is more appropriate if false negatives are costly?

Pearson & deviance residuals

Type	Formula
Pearson	$r_i^P = (y_i - \hat{p}_i) / \sqrt{\hat{p}_i(1 - \hat{p}_i)}$
Deviance	$d_i = \text{sign}(y_i - \hat{p}_i) \sqrt{2 y_i \ln \frac{y_i}{\hat{p}_i} }$

Large absolute Pearson or deviance residuals flag poorly fitted observations.

Hosmer–Lemeshow test

Groups observations into $g = 10$ deciles of predicted probability and compares observed vs. expected counts:

$$H = \sum_{k=1}^g \frac{(O_{1k} - E_{1k})^2}{E_{1k}} + \frac{(O_{0k} - E_{0k})^2}{E_{0k}} \sim \chi_{g-2}^2$$

Non-significant p -value suggests adequate fit.

Pearson residuals via sampled data

```
# Collect a 5% sample for diagnostics
diag_df <- ml_predict(lr, okc_train) %>%
  select(not_working, prediction) %>%
  sample_frac(0.05) %>% collect()

diag_df <- diag_df %>%
  mutate(
```

Logistic Regression — Additional Diagnostics III

```
p_hat = prediction,
pearson = (not_working - p_hat) /
  sqrt(p_hat * (1 - p_hat))
)

# Flag large residuals
diag_df %>%
  filter(abs(pearson) > 3) %>%
  nrow()

# Count of potential outliers
```

Hosmer–Lemeshow test (via ResourceSelection)

```
# Run on sampled data in R
library(ResourceSelection)
hl <- hoslem.test(
  diag_df$not_working,
  diag_df$p_hat,
  g = 10)
hl$p.value
```



```
#  $p > 0.05$  --> adequate fit
```

Influence: Cook's distance for logistic

Collect a sample, fit a base-R `glm()` model, and use `cooks.distance()` or `influence.measures()` to identify influential cases.

- Exponentiate log-odds $\rightarrow$ interpretable odds ratios
- Diagnostics: residual plots, VIF, Cook's D , HL test

sparklyr + Apache Spark MLlib

therinspark.com · spark.posit.co

Luraschi, Kuo & Ushey (2019). Mastering Spark with R. O'Reilly.